

ROTAX 912 iS / SPORT OPERATORS MANUAL (OM)

****Document Ref: OM-912iS-ED02****

****Flight Operations, Pre-Flight Run-Up, In-Flight Checklists & Envelope Limitations****

1. PRE-FLIGHT RUN-UP & INTEGRITY CHECKS

1. ****Engine Warm-Up:**** Run engine at 2,500 RPM until oil temperature reaches minimum 50 deg C (122 deg F).
2. ****Ignition & Lane Cross-Check (4,000 RPM):****
 - * Select Lane A OFF -> verify Lane B RPM drop < 180 RPM, Delta EGT < 40 deg C.
 - * Select Lane B OFF -> verify Lane A RPM drop < 180 RPM, Delta EGT < 40 deg C.
3. ****Full Power Static Check (Take-Off):**** Advance throttle to 100% TPS -> Verify static RPM $\geq$ 5,100 RPM, MAP $\geq$ 98 kPa, Fuel Flow 24 - 28 L/h, Oil Pressure 3.5 - 5.0 bar.

2. IN-FLIGHT EMERGENCY CHECKLISTS

- * ****In-Flight Engine Restart:**** Airspeed 80-90 KIAS, Master Switch ON, Lane A & B ON, Fuel Boost Pump ON, Throttle 15% (idle forward).
- * ****High CHT (> 135 deg C):**** Reduce throttle to 4,600 RPM, command fuel trim enrichment +12%, verify cowl flap open, descend 3,000 ft into denser ambient air.
- * ****Low Oil Pressure (< 2.0 bar):**** Reduce throttle to 4,200 RPM, monitor oil temperature trend, declare urgency, vector toward recovery waypoint.
- * ****Electrical Bus Voltage Sag (< 12.8 V):**** Shed non-essential avionics and ISR radar payloads, verify battery discharge current < 5 A.